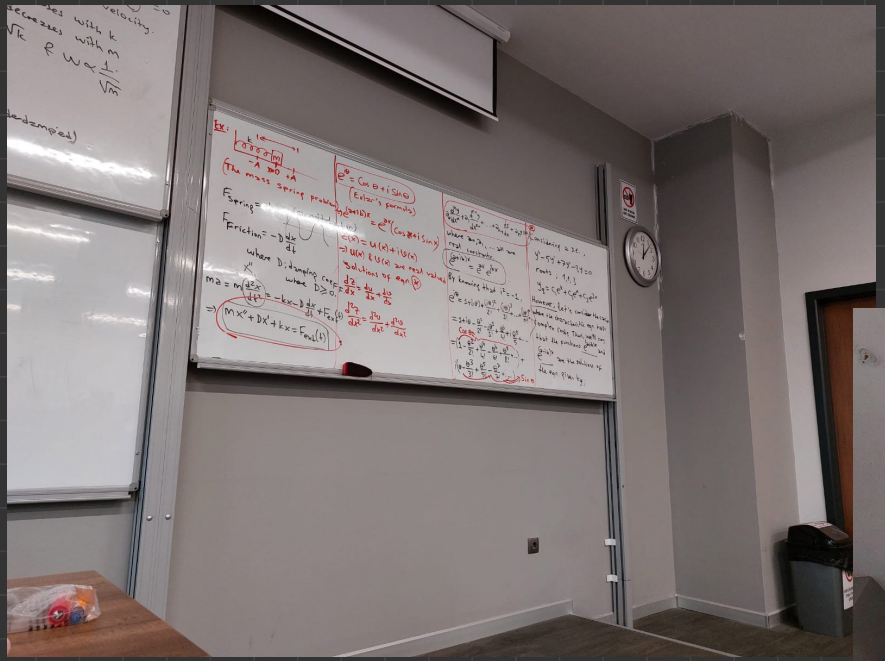
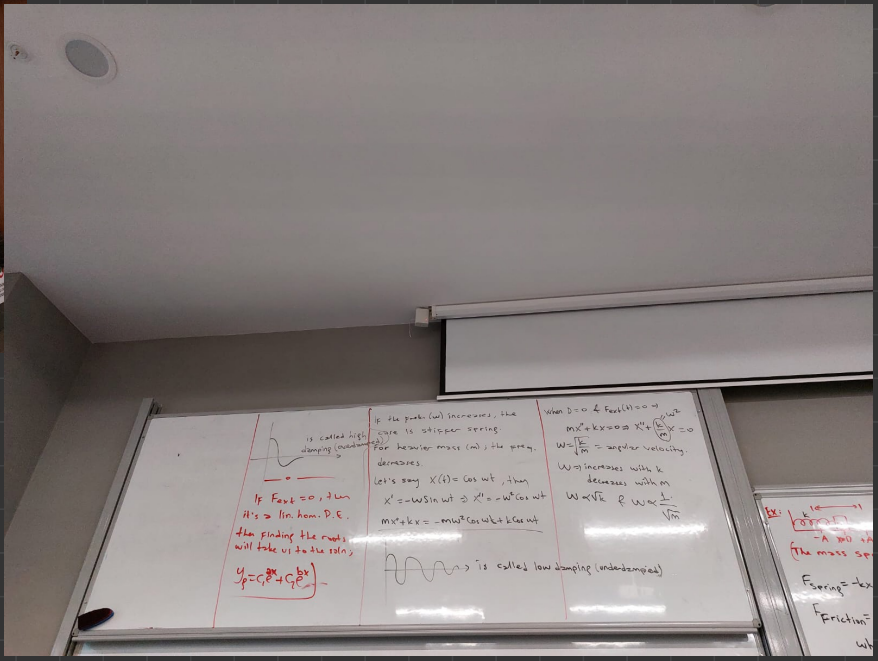
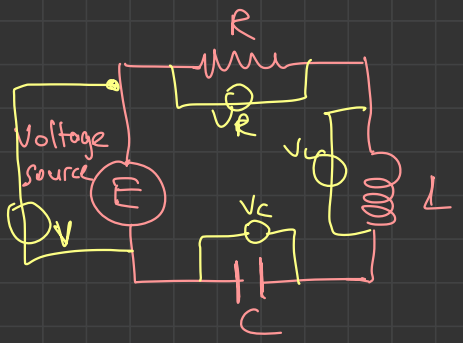

ex:
The mass spring problem
$F_{spring} = -kx$
$F_{Friction} = -D\frac{dx}{dt}$
where D: damping const.
where $D \geq 0$.
$ma = m\frac{d^2x}{dt^2} = -kx - D\frac{dx}{dt} + F_{ext}(t)$
$\Rightarrow mx'' + Dx' + kx = F_{ext}(t)$
$e^{i\theta} = \cos\theta + i\sin\theta$
Euler's formula
$e^{(a+ib)x} = e^{ax}(\cos bx + i\sin bx)$
$z(x) = u(x) + iv(x)$
$\Rightarrow u(x)$ & $v(x)$ are real valued solutions of eqn (*)
$\frac{dz}{dx} = \frac{du}{dx} + i\frac{dv}{dx}$
$\frac{d^2z}{dx^2} = \frac{d^2u}{dx^2} + i\frac{d^2v}{dx^2}$
oscillating s.e.
roots
$y = C_1 e^{\alpha x} + C_2 e^{\beta x}$
$y = C_1 e^{ax}\cos bx + C_2 e^{ax}\sin bx$


is called high damping (overdamped)
If $F_{ext} = 0$, then it's a lin. hom. D.E.
then finding the roots will take us to the soln's
$y_p = C_1 e^{\alpha x} + C_2 e^{\beta x}$
If the prob. (w) increases, the wire is stiffer spring.
For heavier mass (m), the prob. decreases.
let's say $X(t) = \cos wt$, then
$X' = -w\sin wt \Rightarrow X'' = -w^2\cos wt$
$mx'' + kx = -mw^2\cos wt + k\cos wt$
When $D = 0$ & $F_{ext}(t) = 0$
$mx'' + kx = 0 \Rightarrow x'' + \left(\frac{k}{m}\right)x = 0$
$w = \sqrt{\frac{k}{m}}$ = angular velocity.
$w \Rightarrow$ increases with k
decreases with m
$w \propto \sqrt{k}$ & $w \propto \frac{1}{\sqrt{m}}$
is called low damping (underdamped)

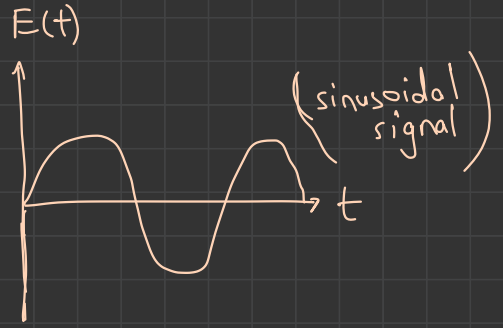
Ex | (Electrical System)

RLC circuit system;



$$E = E_L + E_R + E_C$$

$$V = V_L + V_R + V_C$$



$$E_C = V_C = \frac{q}{C}$$

$$E_R = V_R = I \cdot R$$

$$E_L = V_L = L \cdot \frac{dI}{dt}$$

$\Longrightarrow$ (Kirchoff's Voltage Law)

$$E(t) = L \cdot \frac{dI}{dt} + R \cdot I + \frac{1}{C} \cdot q$$

$$\frac{dE}{dt} = L \frac{d^2I}{dt} + R \frac{dI}{dt} + \frac{1}{C} \frac{dq}{dt}$$

} Diff. Eq with 2nd degree

For ex; $L = 0{,}001\ H$, $R = 0{,}02\ \Omega$, $C = 2F$, $E(t) = \sin 100t$

$$\Rightarrow 0{,}001 \frac{d^2I}{dt^2} + 0{,}02 \cdot \frac{dI}{dt} + 0{,}5 \frac{dq}{dt} = 100 \cdot \cos 100t$$

$$\Rightarrow \underbrace{\frac{d^2I}{dt^2} + 20 \frac{dI}{dt} + 500 \frac{dq}{dt}}_{=0 \text{ for homogenous part}} = 100000 \cdot \cos 100t$$

$r^2 + 20r + 500 = 0 \Rightarrow$ the roots are complex

$(r+10)^2 + 20^2 = 0 \qquad r_{1,2} = -10 \mp 20i$

Hom. part solution $\Rightarrow$

$$I_h(t) = c_1 \cdot e^{-10t} \cdot \cos 20t + c_2 \cdot e^{-10t} \cdot \sin 20t$$

For I_p (particular sol.), we can use the method of undetermined coef.; Thus;

$$I_p(t) = A \cdot \cos 100t + B \sin 100t$$

Then I' and I'' are obtained and substituted into eq.

Then; $A = -10{,}080$, $B = 2{,}122$ are found, which means that

$$I_p(t) = I_p = -10{,}080 \cos 100t + 2{,}122 \cdot \sin 100t$$

$I_{general} = I_h + I_p$

$I_h \rightarrow$ transient current which goes 0 in time.

$I_p \rightarrow$ sinusoidal current (steady state cur) which remains

The Operator Method

$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} \dots + a_n \cdot y$ can be written as

$$(a_0 D^n + a_1 \cdot D^{n-1} + \dots + a_n) y = a_0 D^n y + a_1 D^{n-1} + \dots a_n \cdot y$$

And the solution set of

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_n \cdot y = F(x)$$ (where $a_0, a_1 \dots a_n$ are real numbers)

$L(D) = a_0 D^n + a_1 D^{n-1} + \dots + a_n \Rightarrow$

$L(D)^{-1} \cdot F(x)$ is the solution of equation.

So; $L = (D - r_1)(D - r_2) \dots (D - r_n)$

If $\boxed{Ly = 0}$

$L \cdot y = (D - r_1)(D - r_2) \dots (D - r_n) \cdot y = 0$

$(D - r_i) y = 0$ where $i = 1, 2, 3, \dots, n$

OR $\frac{dy}{dx} - r_i \cdot y = 0.$

After integration, we obtain ;

$y = c_i \cdot e^{r_i \cdot x} \Rightarrow y_G = \sum_{i=1}^{n} c_i \cdot e^{r_i \cdot x}$

General solution

If $\boxed{Ly = Q(x)}$

$(a_0 D^n + a_1 D^{n-1} + \dots + a_n) y = Q(x)$

In this form of eq., we first solve homogenous part and then solve the particubr one. To solve it, we need L^{-1} operato (inverse of L)

$\frac{1}{L} \cdot Ly = L^{-1} \cdot Ly = L^{-1} \cdot Q(x)$

$\Rightarrow y = \frac{1}{L} \cdot Q(x) = L^{-1} Q(x)$

$= \frac{1}{(D-r_1)(D-r_2)(D-r_3)\dots(D-r_n)} \cdot Q(x)$

Let $v = \frac{1}{D - r_n} \cdot Q(x)$

$\Rightarrow (D - r_n) v = Q(x)$

$\Rightarrow \frac{dv}{dx} - r_n \cdot v = Q(x) \Rightarrow v = e^{r_n \cdot x} \int e^{-r_n x} Q(x)\, dx$

Then we go through the second transformation as

$v = \frac{1}{D - r_{n-1}} \cdot v = \frac{1}{D - r_{n-1}} \cdot e^{r_n x} \int e^{-r_n x} Q(x)\, dx$

$y = e^{r_1 x} \int e^{(r_2 - r_1)x} \int e^{(r_3 - r_2)x} \dots \int e^{-r_n x} Q(x) (dx)^n$

is the solution.

Ex: Find the general solution of the given D.E. by using the operator method; $y'' + 5y' + 6y = 0$

Solution:

$$(D^2 + 5D + 6)y = 0$$

$$\Rightarrow (D+2)(D+3)y = 0 \Rightarrow D_{1,2} = -2, -3$$

$$y_g = y_h = C_1 e^{-2x} + C_2 e^{-3x}$$

$y_p = C_1 e^{t} + C_2 e^{-t} + C_3 e^{t}$

Ex: Find y_g of $y'' - y = x$ by using the operator method;

Soln: $D(D^2 - 1)y = x \Rightarrow (D+1)(D-1)y = 0 \Rightarrow \boxed{y_h = C_1 e^{-x} + C_2 e^{x}}$

Particular Soln: $(D-1)u = x \Rightarrow u = \frac{1}{D-1}\, x \Rightarrow \boxed{\frac{du}{dx} - u = x} \Rightarrow u = e^{x}\int e^{-x} x\, dx$

$\Rightarrow \int e^{-x} x\, dx = -x e^{-x} + \int e^{-x} dx = \boxed{-x e^{-x} - e^{-x}}$

$\Rightarrow u = e^{x}(-x e^{-x} - e^{-x}) = -x - 1 \Rightarrow \boxed{u = -(x+1)}$

$dv = e^{-x}$, $w = x$, $v = -e^{-x}$, $dw = dx$

$\Rightarrow y_p = \frac{1}{D+1}\left[-(x+1)\right] = -\frac{1}{D+1}(x+1) \Rightarrow \frac{d^2y}{dx} - dy = 0$

$\Rightarrow (D+1)y = -(x+1) \Rightarrow \frac{dy}{dx} + y = -x - 1$

$\Rightarrow y_p = -e^{-x}\int e^{x}(x+1)dx \Rightarrow y_p = -x$

$y_g = y_h + y_p = C_1 e^{x} + C_2 e^{-x} - x$

Ex

Find the general sol. of $y'' - 7y' + 10y = 3e^x$

$(D^2 - 7D + 10)\,y = 3e^x$

$(D-2)(D-5)\,y = 3e^x$

$\Rightarrow y = \dfrac{1}{(D-2)(D-5)} \cdot 3e^x$ is the solution

$v = \dfrac{1}{D-5}\,3e^x \Rightarrow (D-5)\,v = e^x$

$\Rightarrow \dfrac{dv}{dx} - 5v = e^x$

$\Rightarrow v = e^{5x}$

$(D-2)\,y = 3e^{5x} \Rightarrow y = 3\,\dfrac{e^{5x}}{D-2}$ (call this θ)

$\Rightarrow \theta = \dfrac{1}{D-2}\,e^{5x} \Rightarrow \dfrac{d\theta}{dx} - 2\theta = e^{5x}$

$\Rightarrow \dfrac{d\theta}{dx} - 2\theta = 0$ $\quad \ln\theta = 2x$, $\theta = e^{2x}$

In summary: $(D-2)(D-5)y = 3e^x \Rightarrow y = \dfrac{1}{(D-2)(D-5)} \cdot e^{3x} \Rightarrow v = \dfrac{1}{D-5} \cdot e^x \Rightarrow v = e^{5x}\int e^{-5x}dx = -\frac{1}{4}e^x$

$y = -\dfrac{3}{4}\dfrac{1}{(D-2)}e^x \Rightarrow \dfrac{dy}{dx} - 2y = -\dfrac{3}{4}e^x \Rightarrow y = -\dfrac{3}{4}e^{2x}\int e^{-2x}e^x dx = \dfrac{-\frac{3}{4}e^{2x}\cdot e^{-x}}{(-1)} = \boxed{\frac{3}{4}e^x}$ y_P

$y_G = c_1 \cdot e^{2x} + c_2 \cdot e^{5x} + \dfrac{3}{4}e^x$

$$y'' - 7y' + 10y = 3e^x$$

$$(D^2 - 7D + 10)\,y = 3e^x$$

Homogenous part

$$(D^2 - 7D + 10)\,y = 0$$

$$(D-5)(D-2) = 0$$

$$D = 5,\ D = 2$$

$$y_H = c_1 \cdot e^{5x} + c_2 \cdot e^{2x}$$

Non-homogenous part

$$D = 1$$

$$\frac{1}{D^2 - 7D + 10} \cdot 3e^x \Rightarrow \frac{3 \cdot e^x}{1 - 7 + 10} = \frac{3e^x}{4}$$

Önceki sorunun farklı çözümü

II. Yol

$(D-2)(D-5)y = \varphi(x) = 3e^x \Rightarrow y = \frac{1}{(D-2)(D-5)}\varphi(x)$

$\frac{1}{(D-2)(D-5)}\varphi(x) = \frac{-\varphi}{3(D-2)} + \frac{\varphi}{3(D-5)} = -\frac{1}{3}\left[\frac{\varphi}{D-2} - \frac{\varphi}{D-5}\right]$

$= -\frac{1}{3}\left(e^{2x}\int e^{-2x}3e^x dx - e^{5x}\int e^{-5x}3e^x dx\right)$

$= \boxed{\frac{3}{4}e^x}$

Not: If $Q(x) = e^{rx}$

$$L(D)\, y = e^{rx} \Rightarrow y = \frac{1}{L(D)} e^{rx}$$

$$\Rightarrow y = \frac{1}{L(r)} \cdot e^{rx}$$

Ex

$$y'' + y' + y = e^x \Rightarrow (D^2 + D + 1)y = e^x \Rightarrow L(D) = D^2 + D + 1$$

$$y = \frac{1}{L(D)} e^x \Rightarrow e^{rx} = e^x \Rightarrow r = 1$$

$$\Rightarrow y = \frac{1}{1^2 + 1 + 1} \cdot e^x \Rightarrow y = \frac{1}{3} e^x$$

Ex

$$y'' + 5y' + 4y = 3e^{2x} + 2e^x$$

$$\underbrace{(D^2 + 5D + 4)}_{L(D)} y = 3e^{2x} + 2e^x$$

$$y_h = (D+1)(D+4)y = 0$$

$$y_h = c_1 \cdot e^{-x} + c_2 e^{-4x}$$

$$y_p = \frac{3e^{2x}}{(2^2 + 2 \cdot 5 + 4)} + \frac{2e^x}{(1 + 5 + 4)}$$

$$y_p = \frac{3e^{2x}}{18} + \frac{2e^x}{10}$$

The Power Series Solutions

$$P(x)\,y'' + Q(x)\,y' + R(x)\,y = 0$$

where P, Q and R are the analytic functions.

The Taylor series of a function $y(x)$ around the point x_0 is ;

$$y(x_0) + y'(x_0)(x-x_0) + \frac{y''(x_0)}{2!}\cdot(x-x_0)^2 + \dots = \sum_{n=0}^{\infty} \frac{y^{(n)}(x_0)}{n!}\cdot(x-x_0)^n$$

Ex

$y'' + y = 0$, $y(0)=1$, $y'(0)=0$

$y'' = -y \Rightarrow y''(0) = -y(0) = -1$

$y^{(3)} = -y' \Rightarrow y^{(3)}(0) = -y'(0) = 0$

$y^{(4)} = -y'' \Rightarrow y^{(4)}(0) = -y''(0) = -(-1) = +1$

$$\begin{aligned} y(0) &= 1 \\ y'(0) &= 0 \\ y''(0) &= -1 \\ y'''(0) &= 0 \\ y^{(4)}(0) &= 1 \end{aligned}$$

More explicitely,

$$\Rightarrow \begin{cases} y^{(2k)}(0) = (-1)^k \\ y^{(2k+1)}(0) = 0 \end{cases}$$

where $k = 0, 1, 2 \dots$

continues →

$$\Rightarrow \underbrace{y(0)}_{\parallel 1} + \underbrace{y'(0)\,x}_{\parallel 0} + \underbrace{\frac{y''(0)(x^2)}{2!}}_{\parallel -\frac{1}{2}x^2} + \underbrace{\frac{y'''(0)}{3!}x^3}_{\parallel 0} + \underbrace{\frac{y^{(4)}}{4!}\cdot x^4}_{\parallel \frac{x^4}{24}} \ldots$$

$$\Rightarrow 1 - \frac{x^2}{2} + \frac{x^4}{24} \ldots \quad \Rightarrow \text{The series of } \cos x.$$

$\Rightarrow y = \cos x$ is the unique solution of the given problem.

$$3x + 1 = 4$$
$$x = 3$$

Ex

$$y'' + xy' + (2x+1)y = 0 \quad , \quad y(0) = 1 \, , \; y'(0) = -1$$

$$y'' = -xy' - (2x+1)y \Rightarrow y''(0) = 0 - y(0) = -1$$

$$y''' = -y' - xy'' - 2y - 2xy' - y' \Rightarrow y'''(0) = -(-1) - 2(1) - (-1) = 0$$

$$y^{(4)} = -y'' - y'' - xy''' - 2y' - 2y' - 2xy'' - y'' \Rightarrow y^{(4)}(0) = 7$$

$$\left.\begin{array}{l} y(0) = 1 \\ y'(0) = -1 \\ y''(0) = -1 \\ y^{(3)}(0) = 0 \\ y^{(4)}(0) = 7 \end{array}\right\} \Rightarrow$$

$$y(0) + y'(0)(x) + \frac{y''(0)(x^2)}{2!} + \frac{y'''(0)x^3}{3!} + \frac{y^{(4)}(0)\cdot x^4}{4!} \cdots$$

$$\boxed{1 - x - \frac{x^2}{2} + \frac{7x^4}{4!} + \cdots}$$

Ex

$y'' - xy' - y = 0$ (A), $y(0) = 1$, $y'(0) = 0$

$$y(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n \Rightarrow$$ Let's find out which coefficents should be.

$$a_n = \frac{y^{(n)}(x_0)}{n!}$$ is obtained at the end.

Let $y(x) = a_0 + a_1 \cdot x + a_2 \cdot x^2 + a_3 x^3 \dots$

$y'(x) = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \dots$

$x \cdot y'(x) = a_1 \cdot x + 2a_2 x^2 + 3a_3 x^3 + 4a_4 x^4 \dots$

$y''(x) = 2a_2 + 6a_3 x + 12a_4 x^2 \dots$

$\Rightarrow$ Substitution into eq. (A)

$$y'' - xy' - y = \underbrace{(2a_2 - a_0)}_{0} + \underbrace{(6a_3 - 2a_1)}_{0} x + \underbrace{(12a_4 - 3a_2)}_{0} x^2 = 0$$

$$\begin{aligned} 2a_2 - a_0 &= 0 \\ 6a_3 - 2a_1 &= 0 \\ 12a_4 - 3a_2 &= 0 \end{aligned}$$

$\Rightarrow a_2 = \frac{a_0}{2} \Rightarrow y(0) = a_0 = 1 \Rightarrow a_2 = \frac{1}{2}$

$3a_3 = a_1 \Rightarrow a_3 = \frac{a_1}{3} = 0$ since $y'(0) = a_1 = 0$

$a_4 = \frac{a_2}{4} = \frac{1}{8}$ $\quad a_3 = 0$

$\Rightarrow y(x) = 1 + 0 + \frac{1}{2}x^2 + 0 + \frac{1}{8}x^4$

$\Rightarrow y(x) = 1 + \frac{1}{2}x^2 + \frac{1}{8}x^4$

$$= \sum_{n=0}^{\infty} \frac{1}{2^n \cdot n!} \cdot x^{2n}$$

$$P_n(x) = F(x_0) + F'(x_0)(x-x_0) + \frac{F''(x_0)}{2!}\cdot(x-x_0)^2 + \frac{F'''(x_0)}{3!}(x-x_0)^3 + \frac{F^{(4)}(x_0)}{4!}\cdot(x-x_0)^4 + \dots + \frac{F^{(n)}(x_0)}{n!}(x-x_0)^n$$

$$= \sum_{j=0}^{n} \frac{F^{(j)}(x_0)}{j!}\cdot(x-x_0)^j \quad \text{at point } x_0 \qquad \Rightarrow P_n(x) = \underbrace{F(x_0)}_{P_0''} + \underbrace{P_n'(x_0)}_{F'(x_0)''} + \underbrace{P_n''(x_0)}_{\frac{F''(x_0)}{2!}\cdot(x-x_0)''} + \dots + P_n^{(n)}(x_0) = F^{(n)}(x_0)$$

For example: If $F(x) = e^x \Rightarrow$ around $x_0 = 0$;

$$P_0 = 1,\ P_1 = 1+x,\ P_2 = 1+x+\frac{x^2}{2},\ P_3 = 1+x+\frac{x^2}{2}+\frac{x^3}{6}$$

Not: Taylor polynomial of degree n differs from polynomial of the next lower degree only in the addition of a single term; $P_n(x) = P_{n-1}(x) + \frac{F^{(n)}(x_0)}{n!}(x-x_0)^n$

For $F(x) = e^x \Rightarrow$ If $x_0 = 2$; $e^x \approx e^2 + e^2(x-2) + \frac{e^2}{2!}\cdot(x-2)^2 + \frac{e^2}{3!}\cdot(x-2)^3 + \dots$

For $F(x) = \cos x \Rightarrow$ If $x_0 = 2 \Rightarrow \cos x \approx \cos 2 - \sin 2(x-2) - \frac{\cos 2}{2!}\cdot(x-2)^2 + \frac{\sin 2}{3!}\cdot(x-2)^3 + \dots$

For $F(x) = \sin x \Rightarrow$ If $x_0 = 2 \Rightarrow \sin x \approx \sin 2 + \cos 2(x-2) - \frac{\sin 2}{2!}(x-2)^2 - \frac{\cos 2}{3!}(x-2)^3 + \dots$

Ex

Find the first a few Taylor polynomials approximating the solution around $x_0 = 0$ of I.V.P. ; $y'' = 3y' + x^2 y$, $y(0)=10$, $y'(0)=5$

Solution

$y''(x_0=0) = 3y'(0) + x^2 y(0) = 3.5 + 0 = 15$

$y''' = 3y'' + 2xy + x^2 y' \Rightarrow y'''(0) = 3.15 + 0 + 0 = 45$

$y^{(4)} = 3y''' + 2y + 2xy' + x^2 y''$

$\Rightarrow y^{(4)}(0) = 3.45 + 2.10 + 0 + 0 = 155$

$\Rightarrow P_n(x) = y(0) + y'(0)x + \frac{y''(0)}{2!}(x^2) + \cdots$

$P_n(x) = 10 + 5x + \frac{15}{2}x^2 + \frac{15}{2}x^3 + \frac{155}{24}x^4 + \frac{33}{8}x^5$

$y^{(5)} = 3y^{(4)} + 2y' + 2y' + 4xy'' + 2xy'' + x^2 y'''$

$\Rightarrow y^{(5)}(0) = \underbrace{3.155}_{465} + \underbrace{2.5 + 4.5}_{30} + 0 + 0 + 0$

$= 495$

Ex

Determine the Taylor polynomial of degree 3 for the solution to the I.V.P.

$$y' = \frac{1}{x+y+1}, \quad y(0) = 0$$

Solution

$$y'(0) = \frac{1}{0+y(0)+1} = \frac{1}{1} = 1$$

$$y''(x) = -\frac{(1+y')}{(x+y+1)^2} \Rightarrow y''(0) = -\frac{1+1}{(0+0+1)^2} = -2$$

$$y'''(x) = 2 \cdot \frac{(1+y')^2}{(x+y+1)^3} - \frac{y''}{(x+y+1)^2} \Rightarrow y'''(0) = 2 \cdot \frac{2^2}{1^3} + \frac{2}{1} = 10$$

$$\Rightarrow P_3(x) = y(0) + y'(0)x + \frac{y''(0)}{2!}x^2 + \frac{y'''(0)}{3!} \cdot x^3$$

$$= 0 + 1 \cdot x - \frac{2x^2}{2} + \frac{10}{6}x^3$$

$$\Rightarrow P_3(x) = x - x^2 + \frac{5}{3} \cdot x^3$$

Ex

$$y'' + y' - 2xy = 0$$

Solve the eq. by using the power series solution method

Solution

$$y(x) = \sum_{n=0}^{\infty} a_n x^n$$

Get $y'(x)$, $y''(x)$ and $2xy(x)$ then substitute into eq.

Thus:

$$-2a_0 x + 2a_2 x^0 + a_1 x^0 + 6a_3 x + 2a_2 x + \sum_{n=2}^{\infty} \left[(n+2)(n+1)a_{n+2} + (n+1)a_{n+1} - 2a_{n-1}\right] x^n = 0$$

$$-2a_0 + 6a_3 + 2a_2 = 0$$

$$2a_2 + a_1 = 0 \Rightarrow a_2 = -\frac{a_1}{2} \Rightarrow a_3 = \frac{1}{3}(a_0 - a_2) = \frac{1}{3}\left(a_0 + \frac{1}{2}a_1\right)$$

For $n = 2, 3$;

$$a_{n+2} = \frac{2a_{n-1} - (n+1)a_{n+1}}{(n+2)(n+1)} \Rightarrow a_4 = \frac{\frac{3}{2}a_1 - a_0}{12}$$

$$a_4 = \frac{7}{60}a_0 + \frac{17}{720}a_1$$

$$a_5 = -\frac{3}{40}a_0 + \frac{a_0}{60}$$

Ex

Show that

(Anlamadın, calculus çalışacakmışsın. "Hocan da bilmiyor")

$$x^3 \sum_{n=0}^{\infty} n^2(n-2)\, a_n x^n = \sum_{n=3}^{\infty} (n-3)^2 (n-5)\, a_{n-3} \cdot x^n$$

$$x^3 \sum_{n=0}^{\infty} n^2(n-2)\, a_n x^n = \sum_{n=3}^{\infty} (n-3)^2 (n-5)\, a_{n-3} \cdot x^n$$

Sol:

$\sum_{n=0}^{\infty} n^2(n-2)\, a_n \cdot x^{n+3}$ can be written in this form too.

$k = n+3 \Rightarrow n = k-3$ means that $k = 3$ for $n = 0$

And then replacing k by n

$$\Rightarrow \sum_{n=0}^{\infty} n^2(n-2)\, a_n \cdot x^{n+3} = \sum_{k=3}^{\infty} (k-3)^2 (k-5) \cdot a_{k-3} \cdot x^k$$

So, For some known analytic functions, familiar representations can be written as;

① $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

② $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \cdot x^{(2n+1)}$

③ $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n!} \cdot x^{2n}$

④ $\ln x = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \cdot (x-1)^n$

The formulas given ①, ②, ③ are valid for all x.

But ④ is valid for the interval given in $(0, 2]$

Ex

Find the power series solution of $y' + 2xy = 0$ around $x = 0$

Sol:

$$\sum_{n=0}^{\infty} a_n x^n = y(x) = a_0 + a_1 x + a_2 x^2 + \dots$$

$$y'(x) = a_1 + 2a_2 x + 3a_3 x^2 \dots = \sum_{n=0}^{\infty} n \cdot a_n x^{(n-1)}$$

Substitute y and y' into eq.

$$\sum_{n=0}^{\infty} n \cdot a_n \cdot x^{(n-1)} + \sum_{n=0}^{\infty} 2a_n \cdot x^{n+1} = 0 \quad (\star)$$

$$(a_1 + 2a_2 x + 3a_3 x^2 \dots) + (2a_0 x + 2a_1 x^2 + 2a_2 x^3 \dots) = 0$$

$$a_1 + (2a_2 + 2a_0)x + (3a_3 + 2a_1)x^2 + (4a_4 + 2a_2)x^3 + \dots = 0$$

$$a_1 = 0, \quad 2a_2 + 2a_0 = 0, \quad 3a_3 + 2a_1 = 0, \quad 4a_4 + 2a_2 = 0$$

$a_1 = 0$, $a_3 = 0$, $a_2 = -a_0$

$a_4 = \frac{1}{2} a_0$

$$y(x) = a_0 - a_0 x^2 + \frac{a_0}{2} x^4 + \dots$$

In eq. $(\star)$

$k = n-1 \Rightarrow k = 0$ for $n = 1$, $n = k+1$

$$(\star\star) \Rightarrow \sum_{n=0}^{\infty} n \cdot a_n x^{n-1} = \sum_{k=-1}^{\infty} (k+1)\, a_{k+1} \cdot x^k$$

$$(\star\star\star) \Rightarrow \sum_{n=0}^{\infty} 2a_n x^{n+1} = \sum_{k=1}^{\infty} 2a_{k-1} \cdot x^k$$

for eq $(\star\star\star)$

$k = n+1 \Rightarrow n = k-1$

$k = 1$ when $n = 0$

Continues

$$\Rightarrow \sum_{k=-1}^{\infty} (k+1)\, a_{k+1} \cdot x^k + \sum_{k=1}^{\infty} 2\, a_{k-1} \cdot x^k = 0$$

$$\underbrace{a_1 \cdot x^0}_{=0} + \left[\sum_{k=1}^{\infty} (k+1) \cdot a_{k+1} + \sum_{k=1}^{\infty} 2\, a_{k-1} \right] x^k = 0$$

$$\sum_{k=1}^{\infty} \underbrace{\left[(k+1)\, a_{k+1} + 2\, a_{k-1} \right] x^k}_{=0}$$

$$\Rightarrow a_1 = 0, \; (k+1)\, a_{k+1} + 2 a_{k-1} = 0$$

(recurrence relation)

$$\Rightarrow a_{k+1} = \frac{-2\, a_{k-1}}{k+1} \Rightarrow a_2 = \frac{-2a_0}{2} = -a_0$$

Tüm a değerlerini bulabileceğimiz denklem

Ex Find the general sol. of $2y'' + xy' + y = 0$ around $x = 0$, in the form of power series sol.

$$y(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots$$

$$y'(x) = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 = \sum_{n=0}^{\infty} n \cdot a_n x^{n-1}$$

$$y''(x) = 2a_2 + 6a_3 x + 12 a_4 x^2 + \dots = \sum_{n=0}^{\infty} n(n-1)\, a_n x^{n-2}$$

Substitute into eq.

$$\sum_{n=0}^{\infty} 2n(n-1)\, a_n \cdot x^{n-2} + \sum_{n=0}^{\infty} n \cdot a_n \cdot x^n + \sum_{n=0}^{\infty} a_n \cdot x^n = 0$$

$$\sum_{n=0}^{\infty} 2n(n-1)\, a_n \cdot x^{n-2} \qquad k = n-2 \quad k+2 = n \quad k = -2 \text{ for } n = 0$$

$$= \sum_{k=-2}^{\infty} 2(k+2)(k+1)\, a_{k+2} \cdot x^k$$

$$\sum_{n=0}^{\infty} (n+1)\, a_n \cdot x^n = \sum_{k=0}^{\infty} (k+1)\, a_k\, x^k$$

$$\sum_{k=-2}^{\infty} 2(k+2)(k+1)\, a_{k+2} \cdot x^k + \sum_{k=0}^{\infty} (k+1)\, a_k \cdot x^k = 0$$

$$\left[\sum_{k=0}^{\infty} \Big(2(k+2) \cdot a_{k+2} \Big) + \Big(a_k \Big) \right] \cdot (k+1) \cdot x^k = 0$$

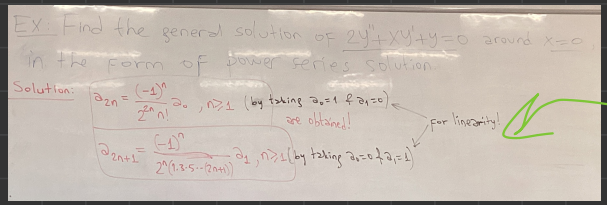
$$2 \cdot (k+2) \cdot a_{k+2} + a_k = 0 \Rightarrow -\frac{a_k}{2(k+2)} = a_{k+2}$$

$$\frac{-a_0}{4} = a_2, \quad \frac{-a_1}{6} = a_3, \quad \frac{-a_2}{8} = a_4, \quad \frac{-a_3}{10} = a_5, \quad \frac{-a_4}{12} = a_6$$

$$y(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \dots$$

$$= a_0 + a_1 x - \frac{a_0}{4} x^2 - \frac{a_1}{6} x^3 + \frac{a_0}{32} x^4 + \frac{a_1}{60} x^5 + \dots$$

$$y(x) = a_0 \left[1 - \frac{x^2}{4} + \frac{x^4}{32} + \dots \right] + a_1 \left[x - \frac{x^3}{6} + \frac{x^5}{60} + \dots \right]$$



BU EN SON !!

For $y(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n$, If $x_0 \neq 0$ then;

Ex

$2y'' + xy' + y = 0$ is given. Find the first few terms in a power series expansion about $x=1$ for a general solution to the given eq.

Sol.

$$y(x) = \sum_{n=0}^{\infty} a_n (x-1)^n \Rightarrow$$ Considering; $x = t+1$

$$Y(t) = y(t+1) = \sum_{n=0}^{\infty} a_n \cdot t^n \Rightarrow \boxed{2\frac{d^2y}{dt^2} + (t+1)\frac{dy}{dt} + y = 0}$$

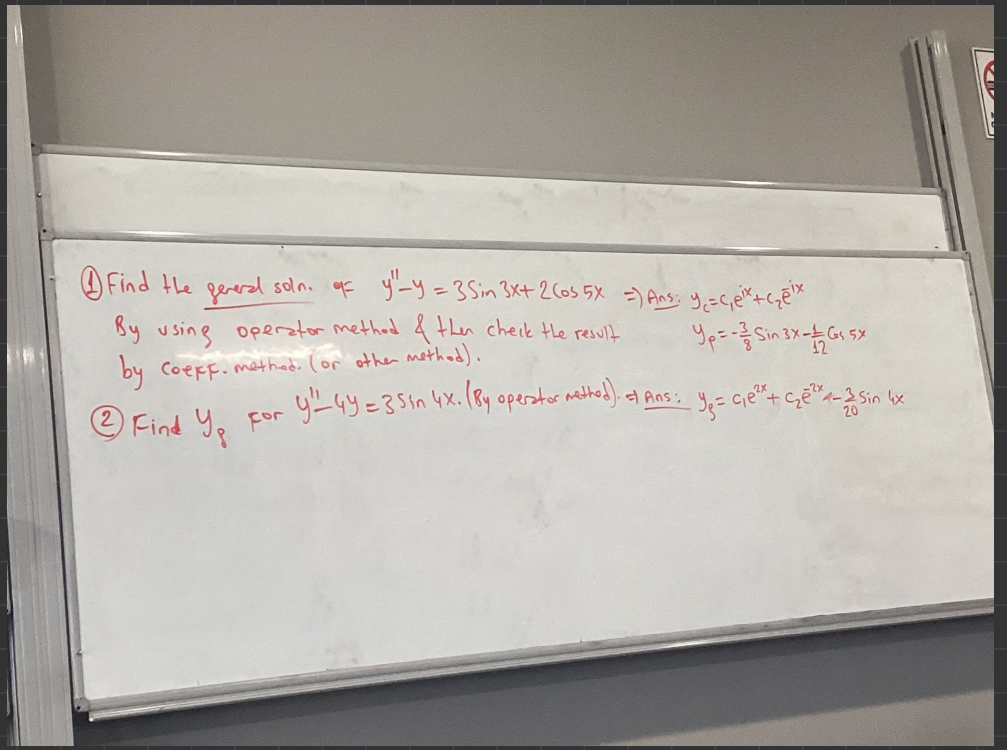
Then we'll get $Y'(t)$ and $Y''(t)$ before substituting them into eq. ✡

Then apply substituting to obtain;

$$Y(t) = a_0\left[1 - \frac{1}{4}t^2 + \frac{1}{24}t^3 + \dots\right] + a_1\left[t - \frac{1}{4}t^2 - \frac{1}{8}t^3 + \dots\right]$$

Then restoring $t = x-1$

$$\Rightarrow y(x) = a_0\left[1 - \frac{1}{4}(x-1)^2 + \frac{1}{24}(x-1)^3 + \dots\right] + a_1\left[(x-1) - \frac{1}{4}(x-1)^2 - \frac{1}{8}(x-1)^3 + \dots\right]$$


① Find the general soln. of $y''-y=3\sin 3x+2\cos 5x$ ⇒ Ans. $y_c=c_1e^{ix}+c_2e^{-ix}$
$y_p=-\frac{3}{8}\sin 3x-\frac{1}{12}\cos 5x$
By using operator method & then check the result by coeff. method (or other method).
② Find y_g for $y''-4y=3\sin 4x$. (By operator method). ⇒ Ans. $y_g=c_1e^{2x}+c_2e^{-2x}-\frac{3}{20}\sin 4x$

Laplace Transform

Suppose that $F(t)$ is defined on the $[0, \infty)$. The Laplace transform of $F(t)$ is determined by,

$$\mathcal{L}\{F\} = F(s) = \int_0^\infty e^{-st} \cdot F(t)\, dt$$

where the Laplace transform of $F(t)$ is denoted by $F(s)$ or $\mathcal{L}\{F\}$.

If the limit exists, then

$$\int_0^\infty e^{-st} F(t)\, dt = \lim_{N\to\infty} \int_0^N e^{-st} \cdot f(t)\, dt$$

Ex Find $\mathcal{L}\{F\}$ of $F(t)=1$ for $t \geq 0$,

$$F(s) = \int_0^\infty e^{-st} \cdot 1 \cdot dt = \lim_{N\to\infty} \int_0^N e^{-st} dt$$

$$= \lim_{N\to\infty} \frac{-e^{-st}}{s} \Big|_0^N = \lim_{N\to\infty} \left[\frac{-e^{sN}}{s} - \left(\frac{-e^0}{s} \right) \right]$$

$$\Rightarrow \lim_{N\to\infty} \left[\frac{1}{s} - \underbrace{\frac{e^{-sN}}{s}}_{\text{zero}} \right] = \frac{1}{s} \qquad \boxed{F(s) = \frac{1}{s}}$$

Ex $F(t) = e^{a \cdot t} \Rightarrow \mathcal{L}\{F\} = ?$ or $F(s) = ?$ (a = const.)

$$F(s) = \int_0^\infty e^{-st} \cdot e^{a \cdot t} dt = \int_0^\infty e^{(a-s)t} dt = \lim_{N\to\infty} \int_0^N e^{-(s-a)t} dt$$

$$= \lim_{N\to\infty} \left[-\frac{e^{-(s-a)t}}{(s-a)} \right]_0^N = \lim_{N\to\infty} \left[-\frac{e^{-(s-a)N}}{s-a} + \frac{e^0}{s-a} \right] = \frac{1}{s-a} = F(s)$$

Ex

Find $\mathcal{L}\{F\}$ for $F(t) = \sin bt$

$$F(s) = \int_0^\infty e^{-st} \cdot \underbrace{(\sin bt)}_{f(t)}\, dt$$

This is given

$$\int_0^\infty e^{au}\left(\sin(nu)\right) du = \frac{e^{au}\left(a.\sin(nu) - n.\cos(nu)\right)}{a^2+n^2}$$

$$\mathcal{L}\{\sin(bt)\}(s) = \lim_{N\to\infty}\left[\frac{b}{s^2+b^2} - \underbrace{\frac{e^{-sN}}{s^2+b^2}\left(s\sin b.N - b.\cos bN\right)}_{0}\right]$$

$$\mathcal{L}\{\sin(bt)\} = \frac{b}{s^2+b^2}$$

Ex

$$F(t) = \begin{cases} 2 \;;\; 0<t<5 \\ 0 \;;\; 5<t<10 \\ e^{4t} \;;\; t>10 \end{cases} \Rightarrow F(s) = ?$$

$$\mathcal{L}\{F(t)\}(s) = F(s) = \int_0^5 e^{-st}.2.\,dt + \underbrace{\int_5^{10} e^{-st}.0.dt}_{0} + \int_{10}^\infty e^{-st}.e^{4t}\,dt$$

$$= 2\underbrace{\int_0^5 e^{-st}dt}_{\text{we found it in prev. questions}} + \int_{10}^\infty e^{-(s-4)t}dt$$

$$= \frac{2}{s} - \frac{2e^{-5s}}{s} + \lim_{N\to\infty}\left[\frac{e^{-10(s-4)}}{s-4} - \frac{e^{-(s-4)N}}{s-4}\right]$$

$$= \frac{2}{s} - \frac{2e^{-5s}}{s} + \frac{e^{-10(s-4)}}{s-4}$$

Theorem - 1 :

① $\mathcal{L}\{F_1 + F_2\} = \mathcal{L}\{F_1\} + \mathcal{L}\{F_2\}$

② $\mathcal{L}\{c \cdot F\} = c \cdot \mathcal{L}\{F\}$

Ex

$$\mathcal{L}\{11 + 5e^{4t} - 6\sin 2t\} = ?$$

$$= \mathcal{L}\{11\} + \mathcal{L}\{5e^{4t}\} + \mathcal{L}\{-6\sin 2t\}$$

$$= 11 \cdot \left(\frac{1}{s}\right) + 5\left(\frac{1}{s-4}\right) - 6\left(\frac{2}{s^2+4}\right)$$

$$= \frac{11}{s} + \frac{5}{s-4} - \frac{12}{s^2+4} \quad (\text{defined for } s > 4)$$

Table of Laplace Transformation

$F(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s)$
1	$\frac{1}{s}$, $s > 0$
e^{at}	$\frac{1}{s-a}$, $s > a$
t^n, $n = 1, 2, 3 \ldots$	$\frac{n!}{s^{n+1}}$, $s > 0$
$\sin bt$	$\frac{b}{s^2+b^2}$, $s > 0$
$\cos bt$	$\frac{s}{s^2+b^2}$, $s > 0$
$e^{at} \cdot t^n$, $n = 1, 2, 3 \ldots$	$\frac{n!}{(s-a)^{n+1}}$, $s > a$
$e^{at} \cdot \sin bt$	$\frac{b}{(s-a)^2+b^2}$, $s > a$
$e^{at} \cdot \cos bt$	$\frac{s-a}{(s-a)^2+b^2}$, $s > a$

Theorem - 2 :

$$\mathcal{L}\{e^{a.t}.\ F(t)\} = F(s-a)$$

Ex $\mathcal{L}\{e^{at}.\ \sin bt\} = ?$

$\mathcal{L}\{e^{at}.\ \sin bt\} = F(s-a) = \frac{b}{(s-a)^2+b^2}$

$\mathcal{L}\{\sin bt\} = F(s) = \frac{b}{s^2+b^2}$

Theorem - 3 :

$$\mathcal{L}\{F'(s)\} = s.\mathcal{L}\{F(s)\} - F(0)$$

Ex Using theorem ③; we can find $\mathcal{L}\{\cos bt\} = ?$

where; $\mathcal{L}\{\sin bt\}(s) = \frac{b}{s^2+b^2}$ is already given

Sol: $F(t) = \sin bt$ and $F(0) = 0 \Rightarrow F'(t) = b.\cos bt$

$\Rightarrow \mathcal{L}\{b.\cos bt\} = s.\underbrace{\mathcal{L}\{\sin bt\}}_{\frac{b}{s^2+b^2}} - 0$

$\Rightarrow b.\mathcal{L}\{\cos bt\} = \frac{s.b}{s^2+b^2} \Rightarrow \mathcal{L}\{\cos bt\} = \frac{s}{s^2+b^2}$

Ex $\mathcal{L}\{\sin^2 a.t\} = \mathcal{L}\left\{\frac{1-\cos 2at}{2}\right\} = \mathcal{L}\left\{\frac{1}{2}\right\} - \mathcal{L}\left\{\frac{\cos 2at}{2}\right\} = \frac{1}{2}\underbrace{\mathcal{L}\{1\}}_{\frac{1}{s}} - \frac{1}{2}\underbrace{\mathcal{L}\{\cos 2at\}}_{\frac{s}{s^2+4a^2}}$

$$= \frac{1}{2s} - \frac{1}{2}\frac{s}{s^2+4a^2}$$

Theorem-4 :

$$L\{t^n \cdot F(t)\} = (-1)^n \cdot \frac{d^n F(s)}{ds^n}, \quad n = 0, 1, 2 \ldots$$

Ex

$$L\{t^2 \cdot \sin at\} = (-1)^2 \cdot \frac{d^2}{ds^2}\left[\frac{a}{a^2+s^2}\right] = \frac{6as^2 - 2a^3}{(s^2+a^2)^3}$$

Ex: $\mathcal{L}\{t^2 \sin at\} = (-1)^2 \frac{d^2}{ds^2}\left[\frac{a}{a^2+s^2}\right] = \frac{6as^2 - 2a^3}{(s^2+a^2)^3}$; $\frac{d^n}{ds^n}$, $n = 0, \pm 1, \pm 2, \ldots$

$= \frac{1}{s}$ $= \frac{s}{s^2+a^2}$

Ex: Solve $y''(t) - y'(t) - 6y(t) = 2$; $y(0) = 1$, $y'(0) = 0$

Soln: $\mathcal{L}\{y''(t)\} - \mathcal{L}\{y'(t)\} - 6\mathcal{L}\{y(t)\} = \mathcal{L}\{2\}$

$\mathcal{L}\{y'(t)\} = s\mathcal{L}\{y'(t)\} - y'(0)$ from theorem ③

By th ③ ; $[s\mathcal{L}\{y\} - y(0)]$

$\Rightarrow \mathcal{L}\{y''\} = s^2\mathcal{L}\{y\} - s\underbrace{y(0)}_{=1} - \underbrace{y'(0)}_{=0} = s^2\mathcal{L}\{y\} - s$

$\mathcal{L}\{y'\} = s\mathcal{L}\{y\} - y(0)$

$\Rightarrow \mathcal{L}\{y'' - y' - 6y\} = \mathcal{L}\{2\} \Rightarrow s^2\mathcal{L}\{y\} - s - [s\mathcal{L}\{y\} - 1] - 6\mathcal{L}\{y\} = 2\underbrace{\mathcal{L}\{1\}}_{=\frac{1}{s}}$

$= (\underbrace{s^2 - s - 6}_{=(s+2)(s-3)})\mathcal{L}\{y\} = \frac{2}{s} + s - 1 \Rightarrow \mathcal{L}\{y\} = \frac{s^2 - s + 2}{s(s+2)(s-3)} = \frac{A}{s} + \frac{B}{(s-3)} + \frac{C}{s+2}$

$\Rightarrow A = -\frac{1}{3}$

$B = \frac{8}{15}$

$C = 1/5$

$\Rightarrow \mathcal{L}\{y\} = Y(s) = -\frac{1}{3}\frac{1}{s} + \frac{8}{15}\frac{1}{(s-3)} + \frac{1}{5}\frac{1}{(s+2)}$

If we transform it,

$\Rightarrow y(t) = -\frac{1}{3} + \frac{8}{15}e^{3t} + \frac{1}{5}e^{-2t}$

4-5 questions

1 → Power Series

1 → Laplace

1 → Operaters

1 or 2 questions → Midterm subjects ⇒

- Substitution
- Cauchy-Eulor
- Reduction of Order
- Exactness
- Variation of parameters
- Linear equations
- Homogenous Eq.